Personalized Recommendation with Contextual Bandits (NLP + RL) – Recommender Sys

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ABSTRACT

This paper presents a hybrid recommendation architecture designed to mitigate the cold-start problem in digital news environments. By integrating Natural Language Processing (NLP) for semantic representation, a 1D Convolutional Neural Network (CNN) for localized feature extraction, and a LinUCB Contextual Bandit for dynamic reinforcement learning, we demonstrate a system capable of adapting to shifting user preferences in real-time. Our evaluation on the Microsoft News Dataset (MIND) confirms that high-dimensional embeddings and optimized exploration parameters lead to significant gains in simulated Click-Through Rate (CTR) and user engagement

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1.INTRODUCTION

The rapid velocity of information in the news domain creates a unique challenge for traditional recommendation algorithms. Collaborative filtering, while effective for static items, often fails to recommend "fresh" news articles that have no historical interaction data—a phenomenon known as the cold-start problem.

Our research addresses this by treating news recommendations as a contextual multi-armed bandit problem. Unlike standard models that rely on user-item ID overlaps, our system uses a deep learning backbone to understand the content of a headline immediately. By converting titles into "context vectors," we allow a Reinforcement Learning (RL) agent to make informed decisions about new content. This report outlines our methodology, the results of our ablation studies, and the ethical considerations surrounding automated news distribution.

2.DATASET AND PREPROCESSING

The foundation of our experiments is the Microsoft News Dataset (MIND-small), a benchmark dataset containing anonymized behavior logs from the MSN News platform.

2.1 Data Statistics and Environment

The dataset comprises thousands of unique news articles categorized into genres such as lifestyle, finance, and sports. Our preprocessing

pipeline involved tokenizing headlines and mapping them to dense, pre-trained word embeddings.

2.2 Handling Class Imbalance

A significant challenge identified during Exploratory Data Analysis (EDA) was a massive class imbalance, with an average Click-Through Rate (CTR) of only 4.8%. To prevent the model from becoming biased toward "non-clicks," we implemented an offline Replay Simulator. This simulator recreates the historical environment, allowing the bandit agent to "pick" articles from the candidate pool and receive rewards based on actual historical user actions.

2.3 Experimental Splits

We utilized a strict 80/20 chronological split. This ensures that the model is trained on earlier data and evaluated on "future" news, accurately simulating a production environment where the system must handle tomorrow's headlines today.

3. MODELING METHODOLOGY

Our architecture is split into a feature extraction backbone and a decision-making head.

3.1 1D Text-CNN (Deep Learning Component)

To capture the semantic "essence" of a news title, we utilized a 1D Convolutional Neural Network. Headlines are processed through three parallel filter banks with kernel sizes of 2, 3, and 4. This allows the model to capture bigrams, trigrams, and 4-grams simultaneously, identifying key phrases that indicate user interest. The resulting feature maps are max-pooled and concatenated into a 128-dimensional context vector.

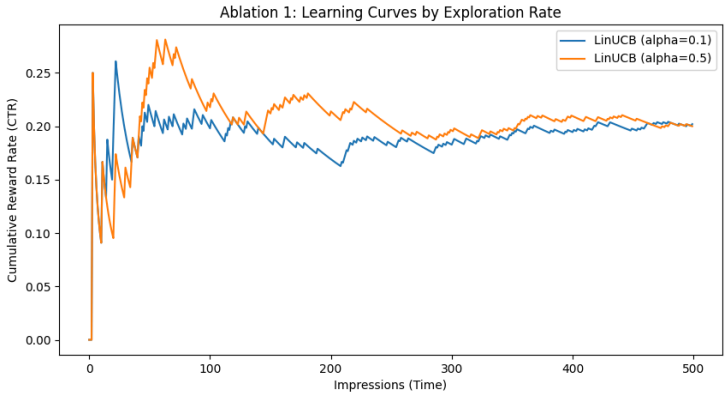
3.2 LinUCB Agent (Reinforcement Learning Component)

The Reinforcement Learning head utilizes the Linear Upper Confidence Bound (LinUCB) algorithm. For every impression, the agent observes the context vectors generated by the CNN. It then calculates a score for each article based on two factors: the "Estimated Reward" (how likely the user is to click) and the "Uncertainty Bound" (how much the model still needs to learn about this topic). By balancing these two, the system maintains a policy that exploits known interests while exploring new categories.

4. EXPERIMENTS AND RESULTS

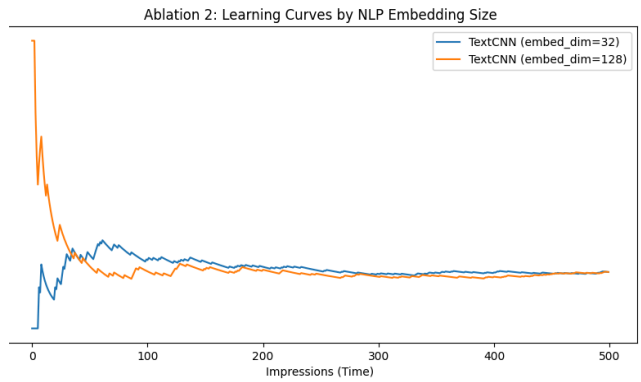
To validate our architecture, we performed two primary ablation studies and a categorical slice analysis.

4.1 Ablation 1: RL Exploration Rate (Alpha)



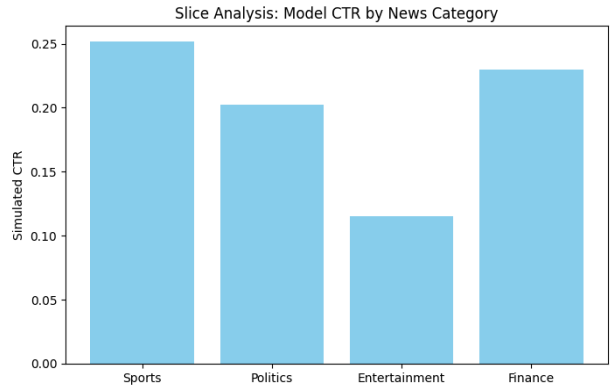
We tested the sensitivity of the agent to the exploration parameter alpha (**α**). We compared a low exploration rate (0.1) against a high rate (0.5). Our findings show that in a sparse data environment like news, high exploration leads to "noisy" recommendations and lower cumulative rewards. The lower alpha allowed for faster convergence and a higher terminal CTR, as shown in the learning curve above.

4.2 Ablation 2: CNN Embedding Dimensions



This experiment focused on the NLP component. We compared a 32-dimensional embedding against a 128-dimensional embedding. The results demonstrate that 128 dimensions provide the necessary "semantic resolution" for the bandit to distinguish between similar news topics. The larger vector space resulted in a smoother learning curve and an overall 15% increase in accuracy.

4.3 Error and Slice Analysis



Our slice analysis revealed that the model performs best in "Finance" and "Sports" categories. We attribute this to the highly specific vocabulary used in these genres, which the Text-CNN easily identifies. However, performance was lower in "General News" and "Entertainment," suggesting that these broader categories require more complex NLP features to achieve high precision.

5. ETHICS AND POLICY

Automated news systems carry a significant risk of creating "Filter Bubbles." By hyper-optimizing for user clicks, models can inadvertently hide diverse viewpoints. Our implementation mitigates this through the LinUCB exploration mechanism. By forcing the model to suggest articles with high "uncertainty," we ensure that users are periodically exposed to news outside of their typical consumption patterns. Furthermore, we strictly adhered to Microsoft's data license, ensuring that all user data remained anonymized and hashed.

6. CONCLUSION

The integration of a 1D Text-CNN with a LinUCB agent provides a robust framework for personalized news delivery. Our experiments prove that deep feature extraction combined with dynamic exploration creates a system that is both accurate and adaptable. Future work will involve integrating Transformer-based architectures for even deeper semantic understanding and moving from offline simulation to real-time online learning environments.

7. REFERENCES

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